

Open set identification of malicious encrypted traffic based on multi-feature fusion

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ARTICLE INFO

Keywords:

SSL/TLS
Malicious traffic detection
Open set recognition
Feature selection
Deep learning

ABSTRACT

In the current network environment, an increasing amount of malicious traffic is transmitted through encrypted channels, carrying control commands and data. With the continuous development of communication protocols and applications, new types of malicious encrypted traffic are emerging, posing significant challenges for network management (e.g., traffic engineering). Therefore, accurately identifying malicious traffic in complex open network spaces has become a hot research topic in network security. In this study, we draw inspiration from channel theory in image science and innovatively convert traffic data into Red-Green-Blue (RGB) image format to achieve the fusion of multiple features. Inspired by image recognition technologies, we have designed a multi-granularity network model that integrates both global and local features, serving as our core network architecture. At the top of the model, we have equipped each known category with a unique autoencoder, using its generated manifold to replace traditional prototypes for model construction. Classification is accomplished through a scoring mechanism that evaluates category membership and by setting thresholds to achieve open set recognition of unknown categories. Relying on our self-created dataset, Malicious and Encrypted Traffic 2024 (MNET2024), we conduct a series of extensive experiments. The results demonstrate that our proposed method exhibits outstanding performance in both closed-set and open-set recognition tasks.

1. Introduction

To ensure the privacy and security of internal data, an increasing number of network communications are now adopting traffic encryption technologies. Widely applied encryption techniques, such as Secure Sockets Layer (SSL), Virtual Private Network (VPN), and DNS over HTTPS (DoS) [1], however, present a challenge to traditional security detection mechanisms. Firewalls, for instance, are incapable of inspecting encrypted traffic, leading to a situation where over 70% of attackers use encrypted protocols to transmit malicious software, conceal their attack activities, and steal sensitive data [2]. As a solution, SSL/Transport Layer Security (TLS) encryption traffic detection technology (also referred to as decryption detection technology) has emerged to fill the gap left by firewalls' inability to scrutinize encrypted traffic [3]. This is achieved by decrypting SSL traffic to inspect encrypted data for potential security threats. Although decrypting SSL traffic is instrumental in uncovering hidden threats and thereby improving network security, the considerable computational resources required for SSL encrypted traffic detection can diminish the throughput performance of firewalls and potentially compromise user privacy.

To protect data transmission and user privacy, most security tools have adopted malicious encrypted traffic identification schemes to

enhance SSL/TLS encryption traffic detection systems. Recent studies indicate that identification of malicious encrypted traffic primarily relies on machine learning methods, such as convolutional neural networks (CNN) and decision trees. These approaches have been effective in detecting known types of malicious encrypted traffic. However, in the real world, due to the continuous development of communication protocols and applications, there exists a significant amount of unknown types of malicious traffic in the network. Traditional neural networks model the entire feature space by comparing the similarity between predicted samples and known category features, leading to classification outcomes. This poses a severe issue: an unknown category that overlaps with known category feature spaces in specific areas is likely to be misidentified as a known class. Consequently, these solutions might incorrectly classify malicious traffic as normal traffic of a known category, resulting in interception failures, and more seriously, allowing malicious traffic to infiltrate the system, causing attacks [4]. Thus, these approaches are no longer sufficient to cope with the complexity of the current network environment.

Although machine learning models have greatly accelerated the process of extracting features from network traffic [5], most existing solutions have not conducted in-depth research on processing raw

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<https://doi.org/10.1016/j.comnet.2024.110824>

Received 30 May 2024; Received in revised form 26 August 2024; Accepted 20 September 2024

Available online 26 September 2024

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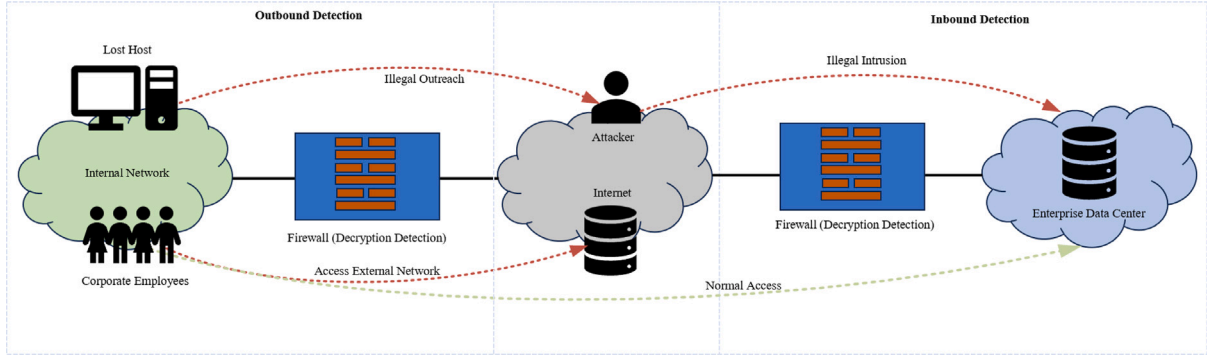


Fig. 1. Firewall traffic detection example. Outbound detection is utilized to prevent internal network clients from malicious software invasions when accessing external websites [8]. When internal network users visit websites compromised with Trojans by attackers, the firewall security system identifies and blocks such malicious traffic to protect the users. Inbound detection, on the other hand, protects the internal network from risks of external access. By deploying firewalls within the web server's network, it ensures that all traffic passing through is secure.

data for malicious traffic identification, still relying on the original traffic data packets or using serialized traffic data as model input. Furthermore, some solutions converted raw traffic into text or images based on specific rules, for instance, Wang [6] mapped the original traffic directly to two-dimensional grayscale images using byte codes. These methods face two major issues: firstly, neural networks generally require input sequences of a fixed length, which may have led to the loss of some information or the introduction of noise; secondly, the abundance of background information make it difficult to extract feature information.

Open set recognition, by training on known samples in the model, can not only accurately classify samples within a closed set space but also identify unknown categories [7]. To enable encrypted traffic detection technology to be applied more efficiently and securely in network management, we propose a new type of malicious encrypted traffic open set recognition model and data processing method. This scheme can monitor traffic in complex network environments and accurately identify known categories of malicious encrypted traffic without decryption. For previously unseen unknown category traffic, our scheme can also timely intercept it and send it to the SSL/TLS encrypted traffic detection system for decryption and detection. This process significantly reduces unnecessary exposure of sensitive data while substantially improving the efficiency of traffic monitoring (see Fig. 1).

Our contributions include:

1. We design a feature fusion method for SSL encrypted traffic. We comprehensively extract sufficient information from the raw data and select specific features from statistical patterns and handshake packet data to enhance the quality of features. Finally, drawing from channel theory in image science, we map three-dimensional features onto the R, G, B channels, forming RGB images as the representation form for model input data.

2. We develop a malicious encrypted traffic open set recognition scheme based on multi-feature fusion. This scheme employs a complex CNN model with multiple branches and feedback mechanisms, capable of extracting both global and local features of images and facilitating their interaction. At the top of the backbone network, we insert unique autoencoders for each known category, implementing open set recognition through the reconstruction of feature semantics and a scoring mechanism.

3. This study conducts extensive experiments based on our self-created dataset MNET2024 and the ISCX VPN-NonVPN dataset. The experimental results demonstrate that our proposed model significantly outperforms several baseline methods.

2. Related work

In this study, we focus on elaborating on the following four categories.

The first category is rule-based pattern matching methods, which represent an earlier approach to identifying malicious encrypted traffic. The open-source intrusion detection system Snort [9] first generates a signature set for sample encrypted traffic and compares it with the signature set of malicious encrypted traffic to determine the category of the sample. Kim [10] and colleagues designed identification rules based on Signatures, Session IDs, and IPs, showing that the identification effect is significantly improved when using multiple rules compared to a single rule. Papadogiannaki et al. [11] used the frequency and location information of packets as rules to design a pattern language describing packets, achieving an identification rate of 84%, indicating that a simple regular expression can possess sufficient expressive power. Husák [12] identified clients through fingerprints in the SSL handshake packets, determining if the traffic was malicious based on the client. However, these methods are not suitable for large-scale traffic identification; moreover, as rules are updated and the signatures of malicious traffic change, the accuracy of identification may also decrease.

The second method involve decrypting and analyzing the traffic. Sourabh Saxena [13] utilized transparent SSL proxy tools such as SSLsplit, stunnel, socat, and ettercap as intermediaries between the client and server. These tools decrypted SSL encrypted traffic and forwarded it, ensuring normal communication between parties while compromising the privacy of the transmitted content.

The third category employed machine learning for traffic identification. Traditional machine learning classification methods included the C4.5 decision tree, random forest, etc. McCarthy et al. [14] utilized four machine learning algorithms (AdaBoost, C4.5, RIPPER, and Naive Bayesian) to learn and evaluate 22 features, demonstrating the feasibility of identifying SSL/TLS encrypted applications using flow-based statistical features and ML algorithms. Shbair et al. [15] proposed a multi-level framework-based HTTPS identification method that did not rely on specific header fields (e.g., easily obtained SNI, TLS certificates, etc.). They used C4.5 and random forest algorithms to train on 18 feature data, such as packet arrival time and packet size from client to server (C2S) and server to client (S2C) directions, achieving a final identification rate of 93.1%. However, a drawback of the C4.5 algorithm is its long training time, making it suitable only for small datasets. Yan et al. [16] introduced the concept of "Burst" by studying the traffic behavior of WeChat red packets. They segmented the traffic into multiple Bursts based on sudden changes in communication bytes and built fingerprints using statistical features such as packet length, TCP handshake counts, inbound and outbound payload, and packet numbers. Experiments comparing K-Nearest Neighbor (KNN), Support Vector Machine (SVM), Random Forest (RF), and GaussianNB (GNB) algorithms showed that the RF algorithm achieved the highest identification rate of 96%, which could further increase to 98% by adding more trees to the RF.

The fourth method utilizes neural networks for traffic identification. Ivan Torroledo and others [17] extracted features from certificate contents and used deep neural networks to determine the validity of malicious certificates, achieving a 94.87% accuracy rate. Blake Anderson and others [18] analyzed traffic statistical features and handshake fields, extracted features such as packet length and byte distribution, and employed logistic regression, decision trees, and other methods for classification training. Lotfollahi et al. [19] proposed a “Deep Packet” traffic identification method, adopting SAE and CNN architectures and selecting 1480 bytes of traffic payload as input, cutting off excess data and padding shorter data to unify dimensions.

With the advancements in machine learning, the field of encrypted malicious traffic identification has witnessed various interdisciplinary attempts. For instance, Wang et al. [6,20] introduced a new traffic pre-processing method, extracting data through Session and Flow forms and converting it into two-dimensional grayscale images for training with a simple CN-ID model. The experiments showed that the Session+ALL combination is the optimal training data form. Wang [21] also explored converting data packets into two-dimensional images for input into CNN models to learn spatial features, then combined with LSTM models to learn temporal features, successfully implementing a real-time intrusion detection system. Chen et al. [22] converted flow data into automatically convoluted images for neural network analysis. Tal Shapira et al. [23] created images from packet sizes and arrival times, named FlowPic, as input for CNN models. Furthermore, deep learning models for text processing, voice recognition, and situational analysis have also been attempted in the encrypted traffic detection field, displaying good performance in their respective domains. For example, the BERT [24] model achieves precise natural language text recognition through contextual processing, addressing the issue of training a large number of model parameters. However, applying images and other applications in the traffic identification domain faces challenges such as: (1) How to solve the problem of excessive background information or incomplete feature selection; (2) How to effectively integrate multi-dimensional features.

3. Data processing methods

By analyzing the structure of SSL/TLS protocol encrypted traffic data packets, this study introduces an innovative method for processing traffic data. Initially, we select and extract traffic features across three dimensions: raw traffic data, handshake packets, and conventional statistical patterns. The characteristics from these three dimensions complement each other, and the three types of features manage to preserve a comprehensive set of characteristics. After standardizing the data, these tri-dimensional feature data are mapped onto the R, G, B channels, transforming them into RGB traffic images to serve as the input data for the model. To balance the number of data samples, we employ supervised data augmentation techniques to expand datasets with fewer samples.

3.1. Data source

To meet the research requirements, we develop the MNET2024 dataset, which includes 10 types of benign encrypted data (such as P2P, web browsing, emails, etc.) and 10 types of malicious encrypted traffic sourced from Stratosphere Lab’s open dataset [25] (like Dridex, Emotet, etc.). For malware with limited sample sizes, we supplement the data by running the respective malware in a safe sandbox environment. To assess the effectiveness of the proposed model, we categorize the 20 types of encrypted traffic into groups of malicious and mixed traffic, randomly assigning unknown categories within the malicious group for experimentation. The specific data distribution is presented in Tables 1 and 2.

Table 1

Benign SSL/TLS encrypted traffic types.

Name	Type	Sample number
Dropbox	P2P	8342
Ebay	Web browsing	7109
Gmail	E-mail	8942
Mega	P2P	9231
NetEase Music	Stream media	7302
Skype	Instant messaging	7843
WhatsApp	Instant messaging	7681
Youtube	Stream media	6993
Instagram	Stream media	6671
News Break	Web browsing	5619

Table 2

Vicious SSL/TLS encrypted traffic types.

Name	CTU number	Sample number
Bunitu	140–1	7831
CCleaner	320–2	7457
Dridex	346–1	7210
Emotet	264–1	8002
HPEMOTET	317–1	7467
PUA.Taobao	237–1	6901
Razy	274–1	6987
Trickbot	327–2	7013
WannaCry	270–1	8021
Heuristic	317–1	9012

3.2. Feature selection

Hexadecimal Traffic Data Research indicates that the valid content of each Packet Capture (PCAP) packet, when output in hexadecimal form, displays distinct characteristics, with the majority of effective bytes falling within the [50,1480] byte range [26]. We read the original binary data stream, store it in an array in hexadecimal format, and then map these data directly to channels to generate images without the need for excessive processing of the original data. According to studies by Wang et al. [6], the overall effect of using Session is superior to Flow under various conditions, though the differences in results are not consistent. Experiments utilizing the ALL layer yielded better outcomes than those using only the L7 layer. Based on these findings, this study ultimately opts for the Session+ALL traffic representation method, selecting traffic from all layers within the same session as the original data stream.

Features sourced from SSL/TLS handshake fields. By analyzing data packet fields with Wireshark, we discover that different SSL/TLS handshakes contain distinct key information. For instance, the cipher suites, compression methods, and extension information in Client Hello and Server Hello packets, as well as the certificate information in Certificate packets. Typically, to reduce attack costs, malicious software families opt for free digital certificates, such as those unconditionally provided by the Let us Encrypt certification authority. Certificate data can reveal extensive information related to malicious software families. Therefore, certain handshake field values in malicious software traffic exhibit significant differences from normal conditions, playing a crucial role in identifying malicious encrypted traffic. As shown in the Table 3, we extract fifteen-dimensional features from the handshake fields.

Traffic Statistical Features The training data for machine learning models primarily utilize preprocessed statistical features because they can mathematically articulate the characteristics of traffic accurately. Thus, this study selects traditional statistical features as the third dimension. Moore and others [27] identified 248 statistical features of common traffic, encompassing a variety of aspects related to ports, time, and space; additionally, Kim and others [28] highlighted that TCP flags, such as the count of data packets with PUSH, Explicit Acknowledgment (ACK), Reset the connection (RST) flags set to 1, serve as crucial features. Specifically, the size of data packets

Table 3
Handshake field feature extraction.

ID	Description
1	Ratio of SSL connections that use successful
2	TLS cipher suites supported by the client
3	Mean public key length
4	TLS cipher suites chosen by the server
5	Parameters for server-side key exchange method
6	Parameters for client-side key exchange method
7	SSL and connection record ratio
8	Number of different certificates
9	Certificate data
10	Mean certificate length
11	Mean number of SAN(Subject Alternate Name)-DNS Domains
12	Indicates if CN in SAN-DNS-domain
13	Indicates if SNI in SAN-DNS-domain
14	Mean duration of certificate validity
15	Standard deviation of the validity period of certificates

Table 4
Flow statistical features.

ID	Description
16	Average duration of the connection
17	Number of PUSH packages
18	Number of RST packages
19	Minimum inter-arrival time of connection
20	Mean inter-arrival time of connection
21	Maximum inter-arrival time of connection
22	Standard deviation of interarrival time between connection
23	Mean secondary time difference of connection
24	Standard deviation of connection's secondary time difference
25	Number of ACK packages
26	Byte count in the forward direction (upstream)
27	Byte count in the backward direction (downstream)
28	Ratio of backward and forward bytes
29	Packet count in the forward direction
30	Packet count in the backward direction
31	Mean forward packet length
32	Mean backward packet length

in malicious traffic often differs from normal traffic. For instance, in Trojan communications, upstream traffic is significantly smaller, whereas downstream traffic is considerably larger than normal. Aiming to control the host, a Trojan's communication patterns differ from the stable characteristics of regular traffic, maintaining communication for extended periods. Based on these insights, we selected seventeen traffic statistical features, as shown in the Table 4.

3.3. Channel fusion

The data standardization process depicted in Fig. 2 encompasses three pivotal modules: traffic segmentation, address randomization, and iterative normalization. The process unfolds as follows. Initially, encrypted traffic is divided by Sessions while removing empty data packets and User Datagram Protocol (UDP) data packets; next, the Media Access Control (MAC) and Internet Protocol (IP) addresses within the data stream are randomized; then, hexadecimal values of multi-dimensional feature data are read two bytes at a time and stored as pixel values in an array; finally, the values within the array are converted to decimals, mapping the range [00, FF] to [0, 255], and data lengths are unified through padding or discarding data, thus completing the standardization.

The publicly available datasets of malicious encrypted traffic are mostly captured at proxies, containing destination MAC and IP addresses that correspond to the proxy tools rather than the actual addresses of malicious software families. Proxy tools are typically applied locally, hence their MAC addresses are local, and IP addresses are often

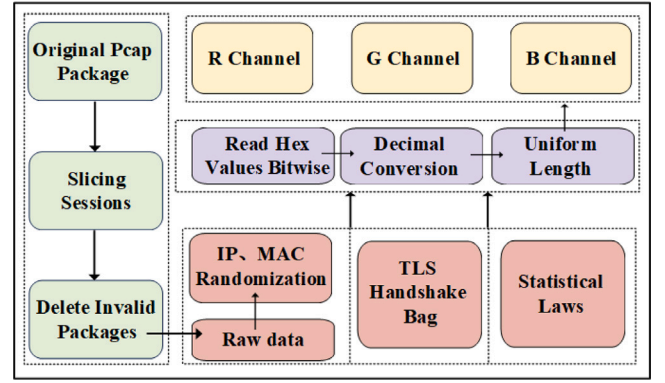


Fig. 2. Data standardization processing module.

127.0.0.1. Due to the fixed values and positions of IP and MAC in the traffic, such highly similar data could lead to model learning biases, making randomization a necessary step.

Normalization refers to mapping data of varying scales to the same scale, ensuring that the processed data has an equal impact on the model. This process is divided into two parts: base conversion and data length unification. RGB pixel values range between 0 and 255, and the decimal conversion range of two hexadecimal values precisely fits this interval, enabling the mapping of all pixel values. In response to the deep learning network's requirements for input data size, data lengths are standardized by truncating excessively long data or padding too short data, thereby unifying the size of images.

The specific technical method for mapping traffic data to channels is as follows. First, we read the content of the PCAP file, where the PCAP file data is represented in binary format. We read the hexadecimal values of the multi-dimensional feature data in two-byte units and store them in an array (this is because we have chosen hexadecimal as the representation of the raw traffic data in one of the three feature dimensions). Finally, for the purpose of value conversion, we convert the hexadecimal values in the array to decimal, i.e., converting the value range [00, FF] to [0, 255], and then reshape the resulting decimal array into a two-dimensional array according to the image width, which gives us the image matrix. Ultimately, we use the Pillow library [29] to save these matrices as Portable Network Graphics (PNG) format image files. Features across three dimensions are sequentially transformed into three grayscale images. Due to variations in data size and output sequence for each feature dimension, the overlapping of these grayscale images produces RGB traffic images that display different morphologies, primarily reflected in the colors of the images. To better segment images of varying granularities, this paper adopts the image dimensions used in pedestrian re-identification [30], settling on a size specification of 128×64 .

From the generated image data, one image per category is randomly selected to compose the visualization effect displayed. The behavior differences among malicious software lead to significant variations between traffic images across categories, especially for the ten types of malicious traffic shown in Fig. 3.

To study the consistency among samples of the same category, four categories of samples are randomly selected, and four images are extracted for comparative analysis. As shown in Fig. 4, despite differences among the four categories, traffic images within the same category generally align.

In summary, traffic images within the same category demonstrate relative consistency, while those between different categories are distinct. Thus, employing image classification methods for encrypted traffic identification in this domain is feasible.

This study supplements the dataset with supervised data augmentation techniques. The imgaug toolkit [31] provides a wealth of data

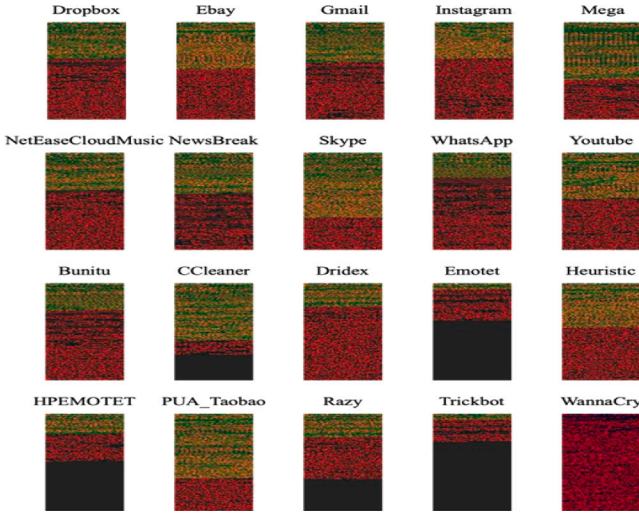


Fig. 3. Visualization results of twenty sample classes.

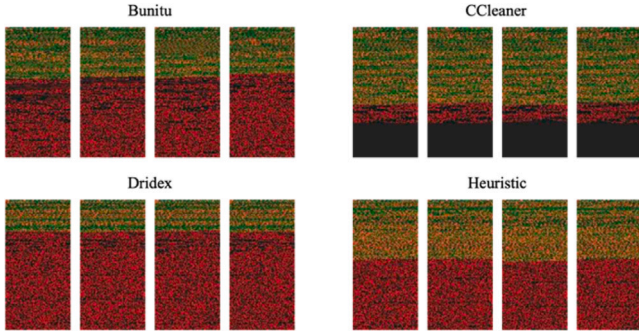


Fig. 4. Consistency analysis of Samples within the same class.

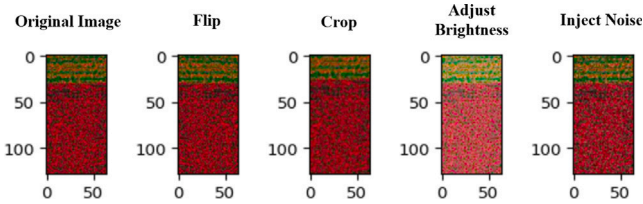


Fig. 5. Renderings of different data enhancement algorithms.

augmentation algorithms. As illustrated in Fig. 5, this research employs four image enhancement methods — image flipping, cropping, brightness adjustment, and noise injection — to augment the image data. It is observed that the processed images across categories remain macroscopically consistent but differ in details, thereby offering valuable sample supplementation for subsequent model training.

3.4. Compared to other methods

In the latest research within the field of traffic identification, deep learning emerges as one of the most frequently utilized methods. Researchers typically need to process the raw traffic into text or images based on specific rules before feeding it into deep learning networks for training and learning. Currently, a common traffic preprocessing approach is to directly map the raw traffic into two-dimensional grayscale images using byte codes. However, there are variations in practice. For instance, Li Ding [32] maintained the positional information in HTTP traffic, including specific markers like the NUL character,

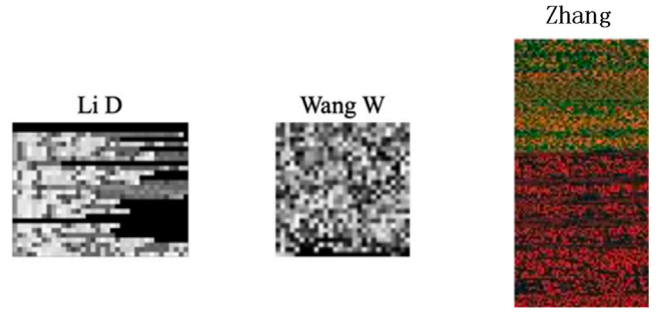


Fig. 6. Effectiveness visualization of the same data augmentation algorithm.

while Wang [6] chose to work with the entirety of the raw data. As depicted in Fig. 6, the distinction between these methodologies is visually discernible. However, both of these strategies harness solely the dimensional features of raw data streams without deeply engaging in the data preprocessing phase, allocating the outcome's quality to the model's capacity.

In contrast, the traffic images generated by this method differ significantly in appearance from the former two. The width of the pictures is roughly the same, but they are longer and are RGB colored images. With the current level of knowledge, it is possible to extract three-dimensional feature data from SSL/TLS encrypted traffic, and the combination of different data in the RGB channels can produce $255 \times 255 \times 255$ different pixel colors. Consequently, mapping the three-dimensional feature data into the three channels generates traffic images that can express traffic characteristics more finely.

3.5. Advantages of regional segmentation in RGB traffic maps

As described in Fig. 7, significant differences exist between Dridex and Heuristic traffic maps. In Dridex traffic maps, the G and B channels feature sparse data, resulting in the lower half of the image predominantly displaying in red and black—red dominated by non-zero data in the R channel, and black indicating zero pixel values. Conversely, in Heuristic traffic maps, only the B channel has relatively sparse data, leading to approximately one-third of the image presenting in red and black. Further comparison reveals that about half of the images in the G and B channels of the Dridex traffic maps are pure black, i.e., zero pixel values, whereas only about half of the images in the B channel of the Heuristic traffic maps are pure black. Dividing the Heuristic traffic map into thirds shows that the last portion of both the G and B channels is pure black.

These color differences between RGB traffic maps reflect the similarities and differences among various traffic types, and regional segmentation of different ratios helps the model capture image features across more dimensions. Thus, RGB traffic maps demonstrate significant advantages in regional segmentation, providing crucial design insights for subsequent model development.

4. Method introduction

Currently, the field of malicious traffic identification has not yet solved the problem of identifying traffic categories outside of predetermined sets, limiting the application of models to specific scenarios. To address this issue, this study designs a multi-granularity network model that integrates both global and local features, and in conjunction with autoencoders, proposes an open set recognition scheme for malicious encrypted traffic. For ease of reference, all the symbols used in this paper are listed in Table 5.

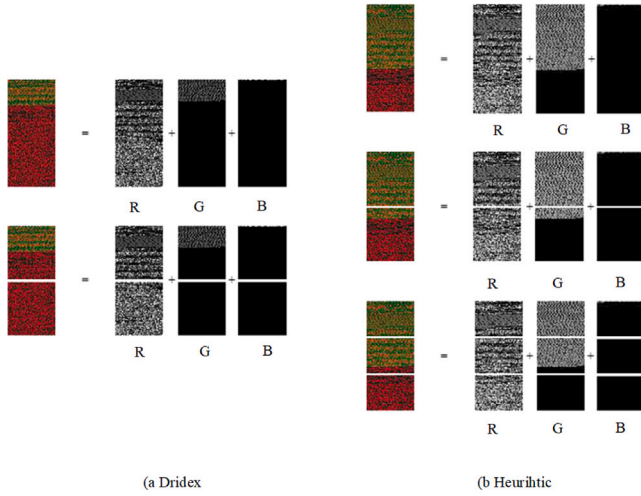


Fig. 7. Example of region segmentation of RGB traffic map. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Table 5
Symbol table.

Symbol	Significance
f	Encoder
g	Decoder
H	Hidden layer
w	A category
E/E_w	Autoencoder/class i-specific autoencoder
p	Pixel-level latent space feature
$\hat{p}$	Reconstructed feature semantic graph
r	Manifold generated by autoencoder
D	Reconstruction error
i	Predicted label
i_x	Class to be predicted
P	Feature semantic map
P^i	A collection of feature maps divided according to predicted categories
λ	Scale Factor
m	The total number of known categories
s^*	Scoring function
$E(s^*)$	About the mean of s^*
$Std(s^*)$	About s^* standard deviation
$w_1 w_2$	Detection score weight

4.1. Prior knowledge

Class prototype methods rely on learning prototypes for each category to achieve open-world recognition. These prototypes are typically a set of representative samples selected from the training data, with each prototype representing a category or cluster. They can be the center of data points (e.g., mean vectors) or specific representative sample points. Classification is conducted by calculating the distance between a sample and each prototype. However, the applicability of this method is limited due to its lack of fitting capability and diversity in prototype representation.

Autoencoders are a type of unsupervised learning neural network formed by a series of encoders f and decoders g , with an internal hidden layer H used to generate a code representation of the input. Autoencoders compress input samples from a high-dimensional space to a low-dimensional space (i.e., the hidden layer H), learning a compressed representation of the original image and then reconstructing the original image, thus possessing excellent feature extraction capabilities. However, a challenge with methods based on autoencoders is that the features learned from reconstructed images might contain irrelevant information, which could adversely affect performance in both closed and open sets (see Fig. 8).

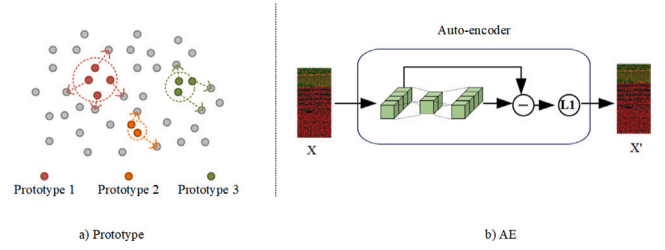


Fig. 8. (a) Prototypical learning. (b) Autoencoder method.

4.2. Reconstruct latent space features

In the method proposed in this study, for each traffic category w , we design a corresponding autoencoder E_w and use E_w to reconstruct the pixel-level latent space feature p extracted by the backbone network, yielding the reconstructed feature $\hat{p}$. The working principle is as follows: the encoder f compresses the high-dimensional semantic feature p into the hidden layer H , learning the mapping from the original feature space to the low-dimensional space $h = f(p)$. The decoder's mapping then generates a manifold $r = g(h)$. We take the latent space features extracted from the backbone network as the input to the autoencoder. This approach, which reconstructs samples at the pixel level rather than directly reconstructing the original images, can reduce the influence of background information (i.e., information irrelevant to the malicious traffic category) on the recognition results, thereby minimizing the reconstruction error as much as possible. The reconstruction error for each pixel is calculated using Formula (1):

$$D(p, E_w) = \|p - E_w(p)\|_1 \quad (1)$$

By calculating the reconstruction error between samples and each category flow, we can classify them. Applying SoftMax to normalize the Logits, the final probability can be defined as shown in Formula (2):

$$p(y = w|p, E) = \frac{e^{-\gamma D(p, E_w)}}{\sum_{j=1}^m e^{-\gamma D(p, E_j)}} \quad (2)$$

where γ is a hyperparameter that controls the hardness of probability distribution. In most prototype learning schemes, the distance between samples and prototypes is calculated using Mean Square Error (MSE). However, evidence [33] shows that MSE may cause the model to overfit, leading to inconsistencies between the prototypes and the actual categories of samples. Therefore, this study adopts Mean Absolute Error (MAE), averaging the predictions for each pixel to make global predictions.

4.3. Unknown category recognition

Autoencoders compress high-dimensional semantic features into a hidden space and minimize reconstruction error, thereby forming a manifold with diverse features. This diversity is reflected in the relationship between categories and features: each category is described by specific parts of the global features.

A sample activates relevant parts of the global features, while irrelevant parts remain inactive. Through the autoencoder, a sample activates features related to its category, which does not result in high reconstruction error; conversely, it leads to significant reconstruction error. This property enables the backbone network to learn to activate only the features related to the category in collaboration with the autoencoder, thereby reducing reconstruction error. The trained model ultimately extracts only features related to specific categories. For unknown categories, the activation rate is low because they are untrained and not associated with any features. Based on the reconstruction error and activation values, this study has established a scoring function. According to the set scoring mechanism, the model sets a threshold

score to ensure that 95% of the known samples are correctly recognized; if a sample scores below the threshold, it will not be recognized as that category. If rejected by all categories, the sample is considered to belong to an unknown category.

Assuming the semantic features P of a given test sample and its predicted label i . As mentioned in Section 4.1, the autoencoder consists of a linear decoder and a linear encoder, satisfying $f(\lambda p) = \lambda f(p)$, and $g(\lambda p) = \lambda g(p)$. Therefore, when constructing a scoring function based on reconstruction error, what needs to be considered is the linear relationship between the latent space features p and the reconstruction error $\|p - E_i(p)\|_1$. For autoencoders with tanh activation, assuming $f(p)$ is close to zero, then $\tanh(\lambda p) \approx \lambda \tanh(p)$. Thus, we have:

$$\|\lambda p - E_i(\lambda p)\|_1 \approx \lambda \|p - E_i(p)\|_1, \quad (3)$$

At the same time, we eliminate the scaling factor λ by dividing by $\|p\|_1$. Additionally, to consider the detection capability of unknown features based on their magnitude, we introduce an extra $\|p\|_1$ term. Thus, the scoring function based on reconstruction error is defined as follows:

$$s_1(p, i) = -\frac{D(p, E_i)}{\|p\|_1^2} \quad (4)$$

Assuming the feature semantic map P_x and the predicted class i_x are obtained from the samples in the training set, we divide the feature maps into different sets based on the predicted categories, defining the feature set $T^i = \{P_x | i_x = i, x = 1, 2, \dots, n\}$, where $i = 1, 2, \dots, m$. In the scoring function based on activation patterns, we primarily consider the average activation level within the same category.

$$\mu_x = \sum_{P \in T^x} \sum_{p \in P} \frac{1}{|T^x| \|P\|} p \quad (5)$$

To also take into account the activation levels of different categories, we will further perform category normalization:

$$\tilde{\mu}_x = \frac{\mu_x}{\sum_j \mu_j} \quad (6)$$

The division of vectors is performed element-wise. To detect unknowns, samples that activate the same features as those activated by known categories are more likely to belong to one of those known categories. Specifically, the activation strength of each feature is weighted by $\tilde{\mu}_i$, and the unknown is assessed based on the weighted average strength of all features. Additionally, we also perform pixel-level score integration. Based on these, the scoring function is formulated as Formula (7):

$$s_2(P, i) = \sum_{p \in P} \frac{1}{|P|} p^\top \tilde{\mu}_i \quad (7)$$

The overall scoring function is derived from normalizing the above two, aiming to standardize the metrics of scoring functions based on different rules by calculating their mean and standard deviation. This process is as Formula (8):

$$\tilde{s}_s(P, i) = \frac{s_s(P, i) - E(s_s)}{Std(s_s)} \quad (8)$$

Finally, this paper combines the three normalized scoring functions through linear combination. The specific scoring process is shown in Fig. 9, and the final overall scoring function is organized as Formula (9):

$$s_{all}(P, i) = w_1 \times \tilde{s}_1(P, c) + w_2 \times \tilde{s}_2(P, c) \quad (9)$$

4.4. Multiple granularity network

The Multiple Granularity Network (MGN), initially proposed by CloudWalk [34] Technology in 2018, is a dynamic target detection model for pedestrian re-identification. Its principal concept involves segmenting an image into multiple parts to extract finer granularity

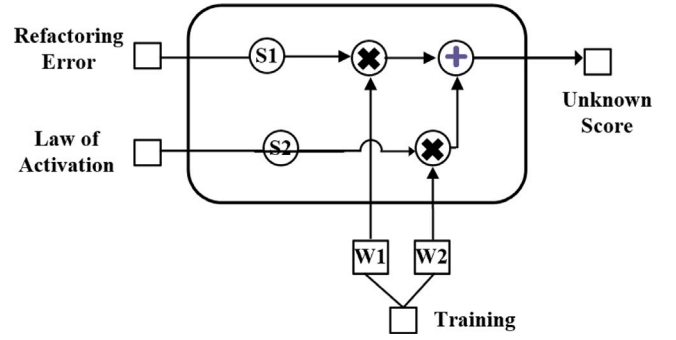


Fig. 9. Scoring mechanism.

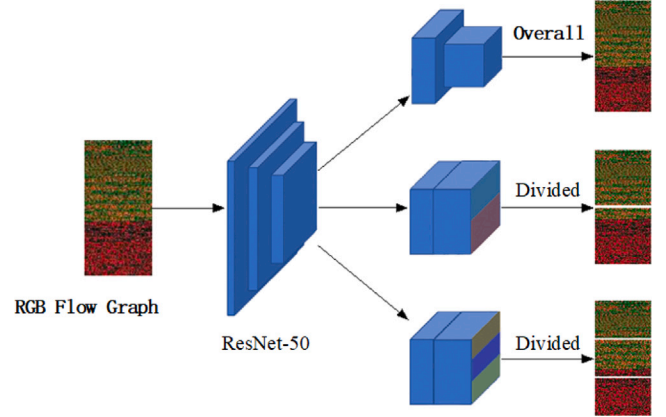


Fig. 10. Basic architecture of multi-branch network. Branch 1 extracts coarse (global) features of the entire traffic map; Branch 2 bisects the traffic map to extract medium-grained features of the upper and lower parts; Branch 3 trisects the traffic map to extract fine-grained features of the top, middle, and bottom parts.

image features. The MGN model has achieved unprecedented success on three open-source ReID datasets: Market-1501, CUHK03, and DukeMTMC-reID, notably reaching a Rank-1 Accuracy of 96.6% on the Market-1501 dataset, which even increased to 97.1% after Re-Ranking, significantly advancing the development of the ReID field.

In RGB traffic images, the pixels at the top often contain features from all dimensions, while the pixels in the lower part of the image may only display features from one to two dimensions. This characteristic allows RGB traffic images and ReID (Re-Identification) pedestrian images to show different levels of feature granularity in various areas, exhibiting a high degree of similarity. Drawing inspiration from the MGN network architecture, we have developed three separate but interconnected branch networks that share a backbone network to extract features of different granularities from the images, thereby obtaining detailed local feature information. These branch networks share the common features extracted by the main network and then conduct a more thorough analysis of the features, achieving recognition of traffic samples from a holistic to a localized perspective.

5. Malicious encrypted traffic open set recognition scheme based on multi-feature fusion

5.1. Model architecture

The multi-branch network model mainly consists of two parts: the backbone network and the branch network. In this paper, the backbone network employs ResNet-50 [35], maintaining the same structure for each layer as its official configuration but only utilizing the first three layers. ResNet-50, a deep network based on the Bottleneck design,

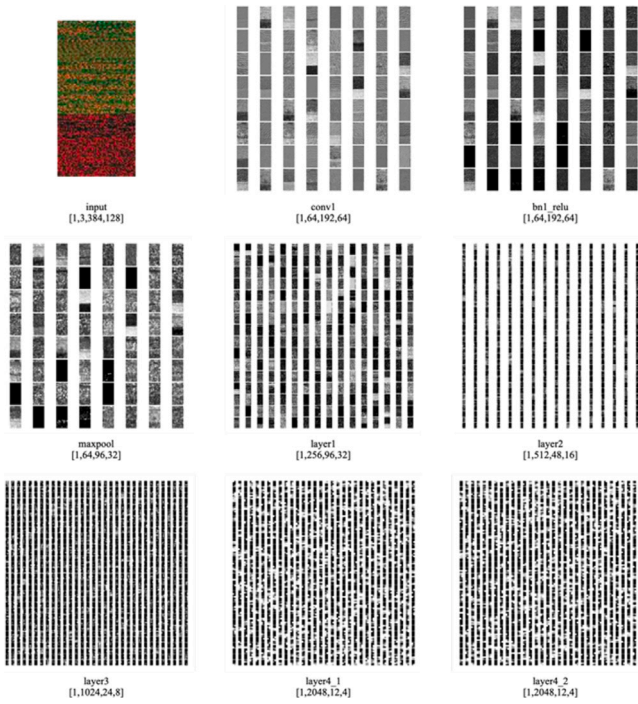


Fig. 11. Visual semantic feature map of backbone network. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

primarily comprises 4 layers, each consisting of several Blocks. Before the first Block, there are two layers of operations: a 7×7 convolution layer and a 3×3 pooling layer, named conv1 and maxpool, respectively. Using Python's PyTorch and Matplotlib tools, this paper visualizes the feature semantic maps output by each layer of the backbone network, as shown in Fig. 11, with four key points to note: Firstly, the input image is a color image based on RGB channels, with a size of $[3, 384, 128]$, meaning it has three channels and a dimension of 384×128 , generated using the method described in Chapter 3; Secondly, the initial convolution layer conv1 uses a 7×7 kernel size, and the pooling layer employs 3×3 maximum pooling, aiming to coarsely extract shallow features of the image, some feature maps show a clear boundary at two-thirds of the way, visually similar to the original traffic map; Thirdly, from layer1 to layer3, each layer contains multiple Blocks, each performing three convolution operations, where the structure between Blocks in each layer is the same but with slight variations in detail parameters, such as the use of stride 1 convolutions in all three layers, but convolutions with stride 2 in layers 2 and 3, aiming to reduce the size of the output image and increase the number of channels, thereby generating more feature maps; Lastly, Fig. 10 includes the feature semantic maps after passing through all the layers of ResNet-50, including the 2048-dimensional feature maps generated by layer4, although layer4 is not used in this model.

The model's other component consists of three branch networks, which operate independently after layer3 of ResNet-50. Each branch undergoes a max pooling process first, with branches 2 and 3 also undergoing local max pooling, followed by a convolutional operation that reduces the feature semantic map from 2048 dimensions to 256 dimensions. Global feature semantic maps directly use Triplet Loss for loss calculation, while local feature maps, after being connected to a fully connected layer, utilize Softmax Loss for loss calculation. Moreover, after the pooling layer in each branch, the global feature maps are additionally fed into a fully connected layer and subjected to Softmax Loss for loss calculation, aiding the model in better learning the global features of images across all branches.

5.2. Scheme introduction

Theoretically, utilizing multidimensional feature data combined with category-specific classifier models is expected to enhance the accuracy of open-set recognition. However, this approach also leads to additional cost burdens, including GPU resources and time consumption, which may not be justified in certain application scenarios.

In response to these challenges, this article focuses on the discussion and design of multiple comparative experiments. Under the premise of reasonable cost control, it evaluates the impact of traditional data processing methods versus multidimensional feature fusion strategies on recognition accuracy. Additionally, it analyzes the efficiency difference between traditional approaches and the category-specific classifier method proposed in this study.

The research first concentrates on the overall design process of the recognition scheme, covering selections from data processing options, backbone network selection, and parameter tuning, to the comparative analysis of final results. Secondly, to ensure the scheme's applicability in various traffic scenarios, a comparison is made using multi-category datasets to specify software configurations and evaluation criteria. Furthermore, based on traffic preprocessing techniques and open-set recognition models, experiments are designed in a unified experimental environment to refine parameter selection. Ultimately, a malicious encrypted traffic open-set recognition scheme based on feature fusion is proposed, compared with existing schemes, showcasing the comprehensive advantages of this approach.

The core architecture of this study is illustrated in Fig. 12. The malicious encrypted traffic open-set recognition process is divided into five major phases: data preprocessing, model training, scheme validation, model evaluation, and scheme deployment.

Compared to other traffic recognition schemes, this proposal guarantees three significant advantages: firstly, it effectively utilizes the multidimensional features of malicious encrypted traffic, including statistical features extracted based on prior knowledge. Secondly, it achieves open-set recognition while ensuring the performance of closed sets. Thirdly, it supports transfer learning, enabling an end-to-end processing flow. The diagram briefly explains the main framework of this scheme. The following sections will detail critical stages such as data preprocessing, model training, and model evaluation.

5.3. Model training

This module encompasses the backbone network ResNet-50, designed for learning latent space features, and specific autoencoders $\{E_i\}_{i=1}^m$ for classifying known classes and detecting unknown classes. The three 256-dimensional semantic feature maps extracted from the network are concatenated into a comprehensive semantic feature map P , which is then input into the specific autoencoder E_i . This autoencoder equally learns and encodes features of all pixels and reconstructs the semantic features of P . The reconstruction error of the class-specific autoencoder is used as Logits, and the logits are multiplied by a coefficient γ to perform pixel-level SoftMax operation. The global prediction is obtained by averaging the predictions of each pixel:

$$p(y = w|P, E) = \frac{1}{|P|} \sum_{p \in P} p(y = w|p, E) \quad (10)$$

Next, the model is trained by minimizing the negative log probability of the true class i using gradient descent, as follows:

$$L = -\log p(y = i|P, E) \quad (11)$$

For the recognition of unknown classes, this scheme uses the pixel reconstruction error derived from the autoencoders corresponding to known classes as input. A scoring mechanism is then employed to assess the probability that an image belongs to an unknown class. After feeding the data into the model, continuous training is conducted to reduce the loss value to the lowest threshold, and then the model is packaged and saved for subsequent model evaluation and scheme deployment.

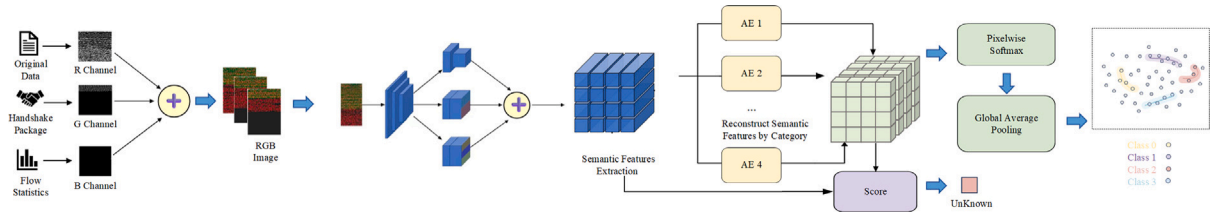


Fig. 12. Model Evaluation Process for Malicious Traffic Detection Based on Object Detection. Original traffic data serves as the input for the entire model, which after data preprocessing, is transformed into RGB traffic maps. These features, extracted by the backbone network for their fundamental semantic traits, are subsequently fed into three branches to capture semantic features of varying granularity. By concatenating the semantic features derived from the three branches, we obtain the ultimate semantic feature map, p . Autoencoders (E_w) for each category w encode and reconstruct these pixel-level semantic features. Following that, the pixel reconstruction error from category-specific autoencoders is used as Logits, with SoftMax operation applied at the pixel level. Then, these pixel-wise predictions are averaged to generate a composite prediction for the entire image. For the recognition of unknown categories, this is implemented by scoring the image's unknown traits.

5.4. Model evaluation

For multi-classification problems, this paper employs macro-averaging or micro-averaging methods [36] to assess the model. Macro-averaging refers to averaging the F1 scores of n binary classification problems, specifically by calculating the Precision and Recall for each of the n binary classifications to obtain n F1 scores, and then averaging these F1 scores to derive the Macro-F1:

$$\text{Macro_F1} = \frac{1}{n} \sum_{i=1}^n F1_i \quad (12)$$

Micro-averaging, however, involves summing the True Positives (TP), True Negatives (TN), False Positives (FP), and False Negatives (FN) of the n binary classification problems to calculate an overall Precision and Recall, thereby obtaining the Micro-F1:

$$\text{Micro_Precision} = \frac{\sum_{i=1}^n TP_i}{\sum_{i=1}^n TP_i + \sum_{i=1}^n FP_i} \quad (13)$$

$$\text{Micro_Recall} = \frac{\sum_{i=1}^n TP_i}{\sum_{i=1}^n TP_i + \sum_{i=1}^n FN_i} \quad (14)$$

$$\text{Micro_F1} = \frac{2 \cdot \text{Micro_Precision} \cdot \text{Micro_Recall}}{\text{Micro_Precision} + \text{Micro_Recall}} \quad (15)$$

The model evaluation process, based on target detection, is illustrated in Fig. 13, where the Gallery Set acts as a database containing all target samples, and the Probe Set is the collection of targets to be recognized. The assessment of model performance is reflected in the overall effectiveness of target recognition within the Probe Set.

First, define the set of images to be detected as the Probe set, containing n images. For each image pr_i in this set, the Probe set can be represented as $Pr = \{pr_1, pr_2, \dots, pr_i, \dots, pr_n\}$. Similarly, define all base images as the Gallery set, containing m images. For each image g_i in the Gallery set, it can be represented as $G = \{g_1, g_2, \dots, g_i, \dots, g_m\}$. We define the number of times a sample of the same type as the image pr_i in the Probe set appears in the Gallery set as K_{pr_i} . The target recognition process is as follows: First, extract the feature vector of the image pr_i from the Probe set and compare it with the feature vectors of all images in the Gallery set, calculating the distance between them. Then, according to the size of the distance, sort the Gallery set in ascending order, and the re-sorted Gallery set is denoted as G_{pr_i} . In this process, if there are matching images for the image pr_i in the Probe set within the Gallery set, then this set of matching images is denoted as $\hat{G}_{pr_i} = \{\hat{g}_1, \hat{g}_2, \dots, \hat{g}_i, \dots, \hat{g}_{K_{pr_i}}\}$. Assume the matching image $\hat{g}_j$ ranks r_j in the re-sorted Gallery set G_{pr_i} and ranks $\hat{r}_j$ in the set of matching images $\hat{G}_{pr_i}$. By repeating this process for all images in the Probe set, we can calculate the overall mAP value of the model as Formula (16).

$$\text{mAP} = \frac{1}{n} \sum_{i=1}^n \sum_{j=1}^{K_{pr_i}} (\hat{r}_j / r_j) \quad (16)$$

When a picture pr_i in the detection set matches the first identified result $\hat{G}_{pr_i}$ (the top-ranked picture after sorting by feature vector distance) as the same type, it is said that pr_i satisfies the first-hit. Let $\hat{n}$

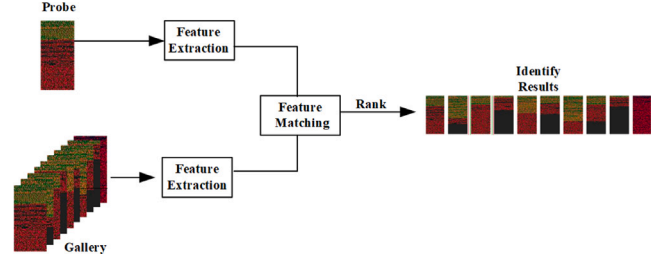


Fig. 13. Model evaluation process based on target detection.

be the number of pictures in Pr that satisfy the first hit, the model's Rank-1 accuracy is defined as Formula (17).

$$\text{Rank} - 1 = \frac{\hat{n}}{n} \quad (17)$$

Correspondingly, the Rank- n value represents the proportion of images hitting within the top n predictions.

This experiment evaluates the effects using accuracy, precision, recall, and F1 score. Specifically, for describing the performance of open-set recognition, we introduce the Area Under the Receiver Operating Characteristic (AUROC) as a threshold-independent metric. This metric is calculated by comparing the True Positive Rate (TPR) with the False Positive Rate (FPR) by varying the threshold. If known and unknown classes are entirely separable, the AUROC value reaches 1.

6. Experiment

The experimental section of this study is divided into three main parts: evaluating the preprocessing methods for SSL/TLS encrypted traffic raw data, comparative analysis with other models, and evaluation of the open-set recognition scheme. For each part, we have designed corresponding experiments to compare with other schemes and have detailed the dataset selection and comparison results.

6.1. Dataset preprocessing methods

This study maps the raw data stream of encrypted traffic, plaintext handshake packet data of TLS, and traditional statistical feature data to the R, G, and B channels, respectively. Furthermore, by selecting three different combinations among the R channel, G channel, B channel, and a blank channel, we generate three types of traffic maps. The blank channel refers to a channel with all-zero data, meaning it contains no data, hereinafter referred to as the N channel (NULL). For instance, Fig. 14 shows the traffic maps of Heuristic type data with different channel combinations, where RNN represents a multichannel map composed of the R channel and two N channels, and RGN is a multidimensional channel map composed of the R, G channels, and an N channel.

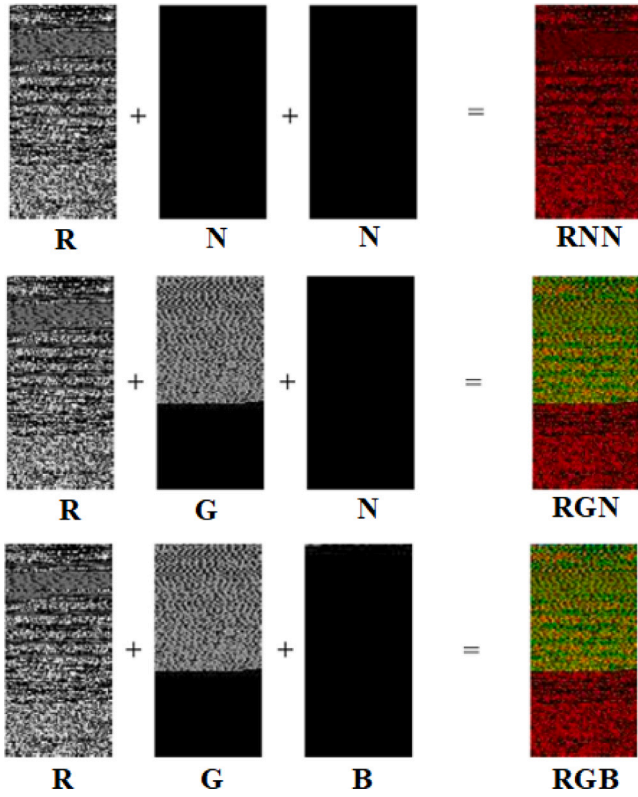


Fig. 14. Illustration of traffic features with multiple dimensions and different combinations.

Table 6
Traffic identification results under different dimensional characteristics.

	Macro-F1	Micro-F1	mAP	Rank-1	Rank-3	Rank-5	Rank-10
RNN	74.86	75.69	91.83	93.22	93.97	94.21	95.78
RGN	76.45	78.46	94.39	95.39	96.64	96.72	97.53
RGB	76.55	78.58	94.42	95.37	96.82	96.98	97.84

In the comparative experiments of this section, the data chosen for model training include the three aforementioned types of traffic maps. For ease of description, we label them as RNN, RGN, and RGB, respectively. The deep learning model uniformly adopts the architecture proposed in this paper, and the evaluation metrics include the micro-averaged F1 value, macro-averaged F1 value, the commonly used mAP value in the target detection field, and the Rank-n hit rate. Table 6 displays the results of traffic recognition under different dimensional features.

According to Table 6, the recognition performance between RGN and RGB does not differ significantly, yet, overall, RGB tends to outperform RGN. This advantage mainly attributes to the additional dimension of statistical feature data incorporated into RGB, enriching the traffic map with more detailed information. However, it is noteworthy that the top-1 hit rate for RGB is slightly lower than that for RGN, a result consistently observed across multiple experiments. This might be due to the visual similarity among certain types of RGB traffic maps, for instance, some WannaCry traffic maps might be misidentified as Dridex type. Moreover, both RGB and RGN significantly outperform RNN, indirectly validating the feasibility and superiority of this scheme.

The analysis above demonstrates that different combinations of multidimensional features of encrypted traffic directly influence the model's recognition performance, with traffic maps utilizing the RGB combination yielding the best results. This underscores the significant impact of data preprocessing on the overall scheme, where choosing

Table 7
The Details of ISCXVPN2016.

Traffic type	Traffic content
Web browsing VPN-Web browsing	Firefox, Chrome
Email VPN-Email	SMTPS, POP3S, IMAPS
Chat VPN-Chat	ICQ, AIM, Skype, Facebook, Hangouts
Streaming VPN-Streaming	Vimeo, Youtube
File Transfer VPN-File transfer	Skype, FTPS, SFTP
VoIP VPN-VoIP	Facebook, Skype, Hangouts, Voice
P2P VPN-P2P	uTorrent, Transmission

and processing the right feature data often leads to substantial improvements. Currently, most researchers simply convert the raw traffic bytecodes directly into grayscale images for training deep learning models, generating images similar to the aforementioned RNN traffic maps. Although this approach minimizes manual intervention, the quality of recognition is entirely dependent on the model design. Thus, leveraging prior knowledge for traffic preprocessing, as in the multidimensional feature extraction process of this scheme, proves invaluable. Experimental results confirm that appropriate training data selection can significantly enhance the model's recognition accuracy.

6.2. Comparison with other models

To demonstrate the characteristics and effectiveness of the aforementioned models in our scheme, this section conducts a comparative analysis of multiple models on the same dataset. The dataset used is the VPN-nonVPN (ISCXVPN2016) dataset, which was open-sourced by Draper-Gil G [37] and others in 2016. The comparative models include C4.5 proposed by Draper-Gil G and one-dimensional CNN proposed by Wang W [6]. In order to differentiate between different models, the notation "model" is used throughout. For example, the model proposed in this paper is denoted as MGN_OS.

Data augmentation techniques are disabled in this study, and Draper-C4.5, Wang-1CNN, and the model proposed in this paper are evaluated on a dataset consisting of 12 classes of traffic, including emails, as shown in the Table 7.

The experimental results, as shown in Fig. 15, indicate that our proposed model performs better overall in terms of precision and recall compared to the C4.5 algorithm used by Draper-Gil G. It even outperforms C4.5 by approximately 23% in the VPN-Chat category. Furthermore, compared to the one-dimensional CNN used by Wang W, our model only achieves slightly lower scores in certain categories such as Streaming and File, but shows a significant improvement overall, especially in the Email and Chat categories.

Additionally, in terms of average precision, Draper-C4.5 achieves 86.4%, Wang-1CNN achieves 89.3%, and our model achieves 97.8%. Regarding average recall, Draper-C4.5 achieves 85.4%, Wang-1CNN achieves 89.8%, and our model achieves 96.5%. Our model exhibits outstanding performance in both metrics.

6.3. Open set recognition performance

To better demonstrate the recognition performance of the proposed method, this section uses the MNET2024 dataset built in this paper for experiments.

A group of experiments is randomly selected, with Razy, Trickbot, and Wannacry as the unknown class in the first group. The traffic recognition rate is shown in the Table 8.

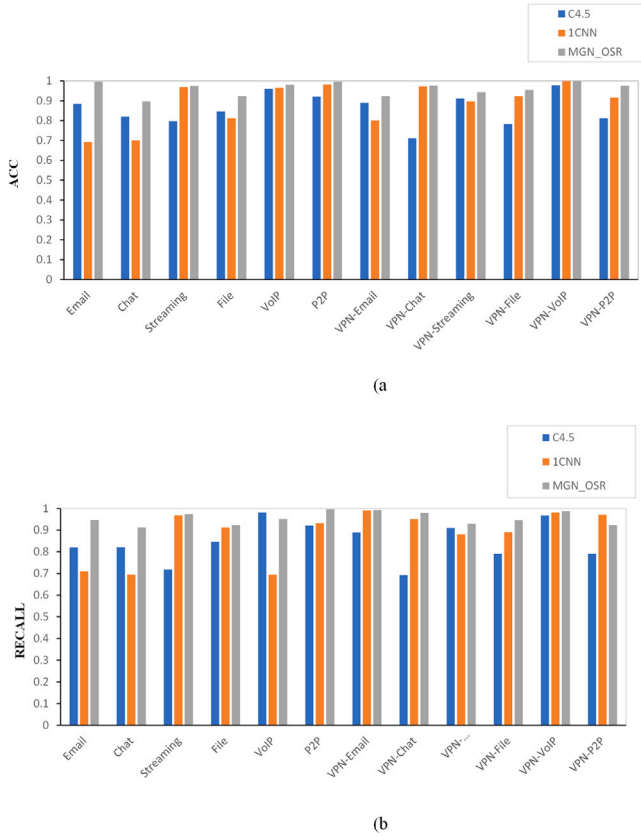


Fig. 15. Results of multi-classification experiments. (a) Accuracy results of multi-classification experiments. (b) Recall results of multi-classification experiments.

Table 8
Traffic identification accuracy.

Name(Benign)	Accuracy	Name(Malicious)	Accuracy
Dropbox	98.2	Bunitu	89.2
Ebay	89.2	CCleaner	96.7
Gmail	96.5	Dridex	99.8
Mega	97.3	Emotet	97.4
NetEase Music	95.2	Heuristic	94.9
Skype	93.7	HPEMOTET	98.9
WhatsApp	98.2	PUA.Taobao	92.4
Youtube	96.4	Razy(Unkwn)	97.6
Instagram	99.5	Trickbot(Unkwn)	95.8
News Break	97.8	WannaCry(Unkwn)	99.3

This paper compares the proposed method with different but related architectures, such as autoencoder-based methods [38–40] and newer class prototype methods [40–42]. The experiments are conducted using the dataset constructed in this paper (MNET2024), VPN-nonVPN (ISCXVPN2016) dataset, and the CTU malware dataset [43]. The experimental results are shown in the 9. Apart from a slight lag behind C2AE on ISCXVPN2016, our model performs better on the other two datasets, especially on the dataset constructed in this paper (MNET2024) with an improvement of 2.3%. This also indicates that our scheme can accurately identify unknown classes while ensuring closed-set performance.

6.4. Comparison with other plans

In comparing different schemes for the recognition of malicious encrypted traffic, researchers have proposed a variety of solutions ranging from superficial features to deep features, from machine learning to deep learning, and from simple network architectures to complex

Table 9
Comparison of AUROC metrics on the unknown class detection task.

Method	MNET2024	ISCXVPN2016	CTU
C2AE [21]	91.3	92.5	83.3
MLOSR [19]	93.7	88.2	82.5
GFROSR [22]	83.1	87.3	80.7
OSRCI [22]	84.2	65.3	73.2
GCPL [23]	87.3	86.8	79.4
PROSER [24]	94.5	87.9	81.2
MGN_OS	96.8	92.1	86.5

ones. With the continuous advancement in technology, the accuracy of these recognition schemes has been steadily improving. When ample physical and time resources are available, selecting the appropriate data format and network model can significantly enhance efficiency. Therefore, this section compares the characteristics of different schemes to further highlight the efficiency and superiority of our approach. For the task of recognizing malicious encrypted traffic, current research has shown that the iterative update of technology from surface feature extraction to deep learning has led to significant improvements in recognition effects. With sufficient computational resources and time, choosing the appropriate data processing methods and network models can greatly improve the efficiency and accuracy of recognition. This section, through a detailed comparison of existing different schemes, further demonstrates the significant advantages of our scheme in terms of efficiency and performance.

Comparative schemes include MARK-ELM (2015) [44], VAEN (2017) [34], HAST-IDS (2018) [21], LeNet-5 (2019) [23], and HexCNN-1D (2023) [26], with the results shown in Table 10.

In the model evaluation phase, our research's data preprocessing stage stands out significantly from other approaches. Our scheme integrates three-dimensional feature data of malicious encrypted traffic, whereas other schemes, such as VAEN and HAST-IDS, rely solely on single-dimensional feature data of traffic, opting only for specific segments of raw data. The LeNet-5 scheme, on the other hand, constructs images based on the size of data packets and their arrival times, without much human intervention in feature engineering. Thus, our approach is unique in extracting multidimensional traffic features. During the model training phase, we utilize RGB traffic maps as the data carrier, extracting multi-granular features of the traffic maps through a multi-branch CNN model based on ResNet-50. The interaction and feedback among features from different branches endow our scheme with the capabilities of multi-granular feature extraction, end-to-end recognition, and transferable learning. In contrast, the HAST-IDS scheme uses both CNN and LSTM deep learning models to extract spatial and temporal features of traffic, illustrating its multi-granular feature extraction properties as well. Furthermore, this paper explores the new direction of open-set recognition, an area not yet extensively studied in the field of malicious traffic recognition.

7. Conclusion and future work

This article proposes an innovative encrypted malicious traffic open set identification scheme based on multi-feature fusion. The scheme deeply explores the entire process from traffic preprocessing to the selection of deep learning models. By converting the original traffic data into RGB images, the scheme not only achieves multidimensional traffic selection and fusion but also effectively eliminates redundant background information, significantly improving the quality of the data input into the model. In terms of model selection, we use a multi-branch structure network to extract and learn features at different granularity levels, making the most of all available features. Finally, by modeling on the manifold generated by the category-specific encoder, we successfully achieve open set identification in the field of traffic recognition. Through extensive experiments on our self-built MNET2024 dataset and

Table 10
A comparison of the characteristics of different schemes.

Program features	MARK-ELM	VAEN	HAST-IDS	LeNet-5	HexCNN-1D	MGN_OSR
Multidimensional feature extraction and fusion	×	×	×	×	×	✓
Multi-granularity vector feature extraction	×	×	✓	×	×	✓
Open set recognition	×	×	×	×	×	✓
Data processing methods	×	×	✓	×	✓	✓

the publicly available ISCXVPN2016 dataset, and by comparing with other research results, the model shows significant recognition effects in both closed and open sets.

To further explore the application scenarios of the model, we identified two issues that need further research in future work. First, although many studies have achieved good results on public datasets, the real samples faced in industrial environments often have a serious imbalance problem, such as a certain type of sample being far more numerous than another type. This data imbalance may lead to problems such as model overfitting, a common challenge in machine learning. In practical applications, it is almost impossible to achieve consistency in the amount of training data samples for all types. Therefore, how to implement reasonable sampling strategies at the data level (to avoid undersampling or oversampling), adjust network parameters at the algorithm level, and use data augmentation techniques to increase the sample size are issues that need further exploration.

Second, with the rapid development of network protocols and applications, the current network environment is becoming increasingly complex. Open set traffic identification is just the first step; future research should also consider how to build models in a network environment that is updated in real-time. Exploring how to use detected unknown category traffic for real-time training of the model, thereby learning from newly emerging malicious or benign traffic to expand the model's recognition capabilities, is also an urgent problem to be solved in future research.

Dataset & code availability

The code and data related to the experiments are available at the provided link: <https://github.com/BeerHan/Code-and-data-for-MGN-O-SR>.

CRedit authorship contribution statement

Xingwen Zhao: Methodology, Funding acquisition. **Han Zhang:** Writing – review & editing, Writing – original draft, Validation, Methodology, Data curation. **Hui Li:** Resources, Funding acquisition. **Xuangui Chen:** Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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